

Distributed Quantum Computing

Massimo Equi

Aalto University

15 June 2026

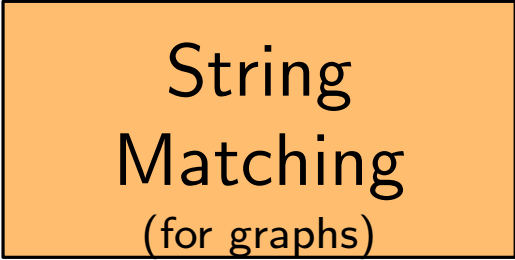
Interview for IQM

About Me

Three main research topics

About Me

Three main research topics

An orange rectangular box with a black border, containing the text 'String Matching (for graphs)'.

String
Matching
(for graphs)

About Me

Three main research topics

String
Matching
(for graphs)

Quantum
Computing

About Me

Three main research topics

String
Matching
(for graphs)

Quantum
Computing

Distributed
Computing

About Me

Three main research topics

String
Matching
(for graphs)

Quantum
Computing

Distributed
Computing

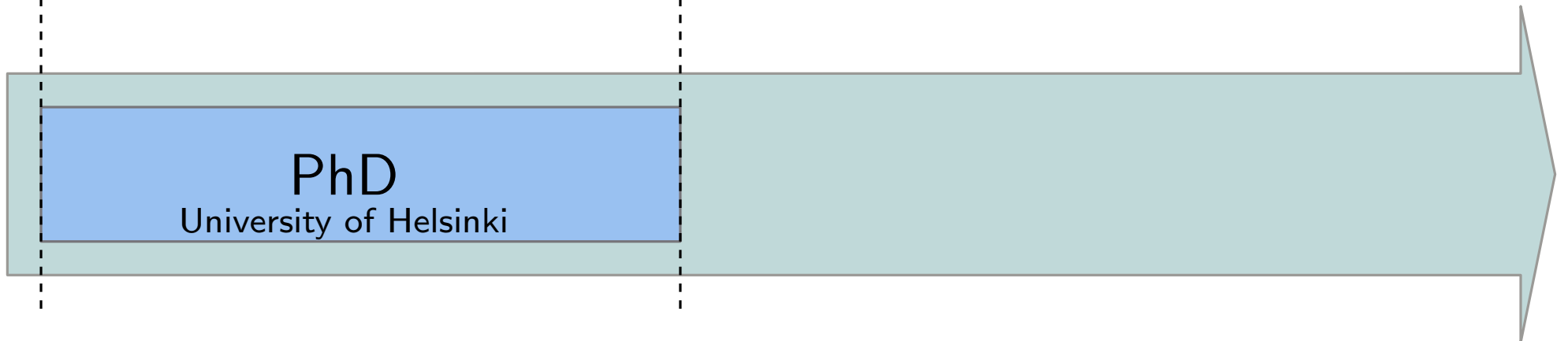
String Matching

PhD

University of Helsinki

2019

2022



About Me

Three main research topics

String
Matching
(for graphs)

Quantum
Computing

Distributed
Computing

String Matching

String Matching

Quantum Computing

PhD

University of Helsinki

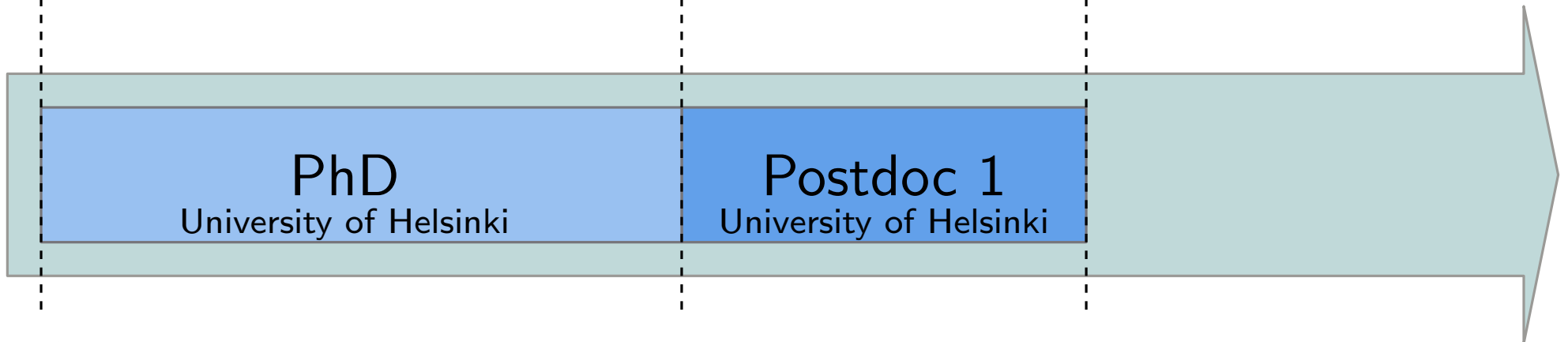
Postdoc 1

University of Helsinki

2019

2022

2024



About Me

Three main research topics

String
Matching
(for graphs)

Quantum
Computing

Distributed
Computing

String Matching

String Matching

String
Matching

Quantum Computing

Quantum Computing

Distributed Computing

PhD

University of Helsinki

Postdoc 1

University of Helsinki

Postdoc 2

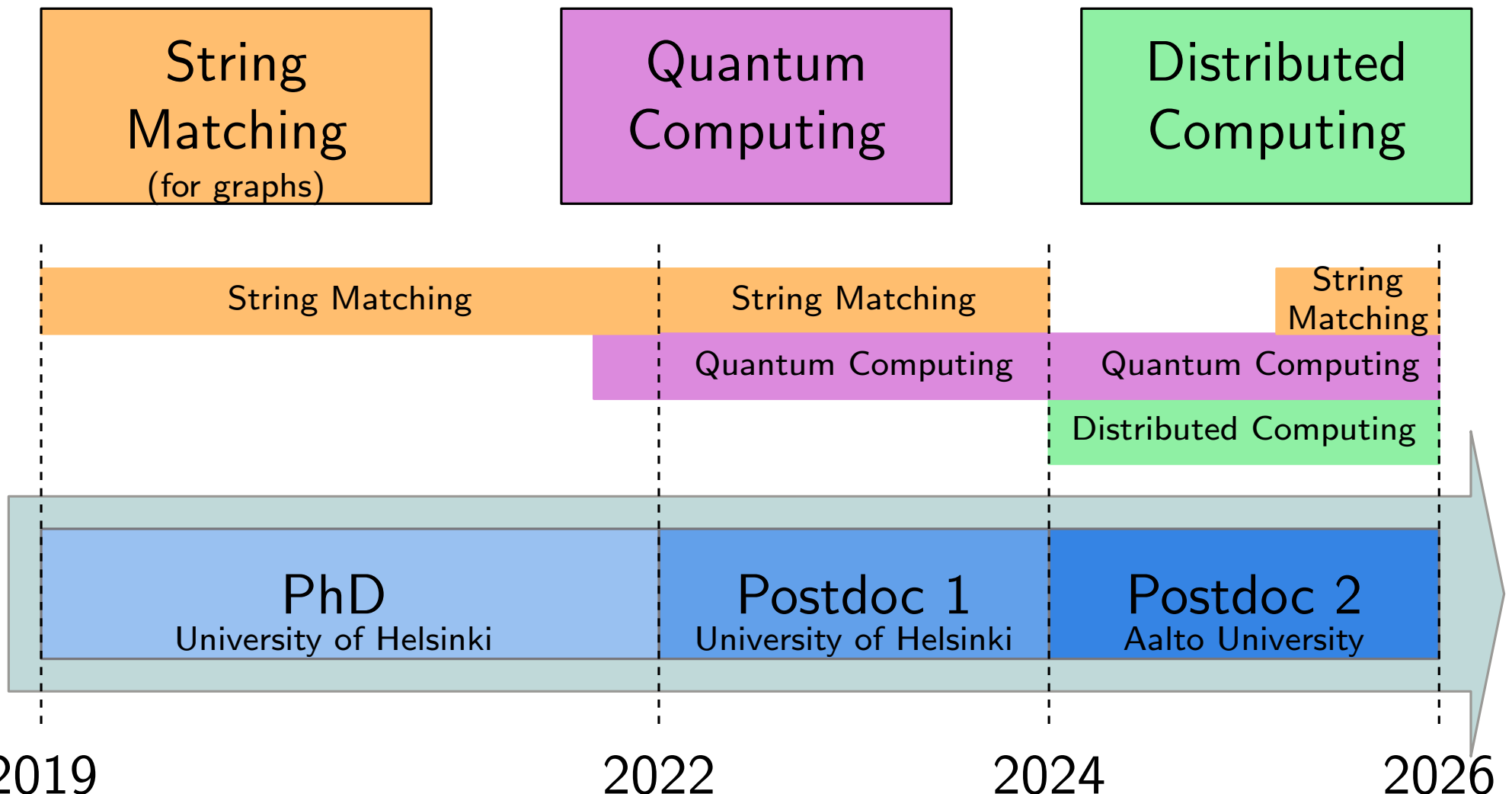
Aalto University

2019

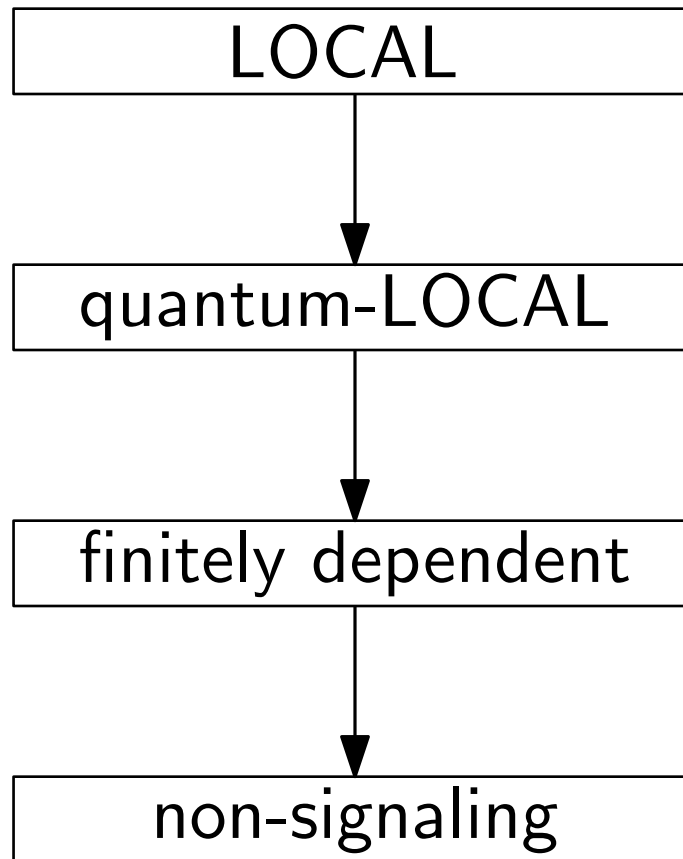
2022

2024

2026

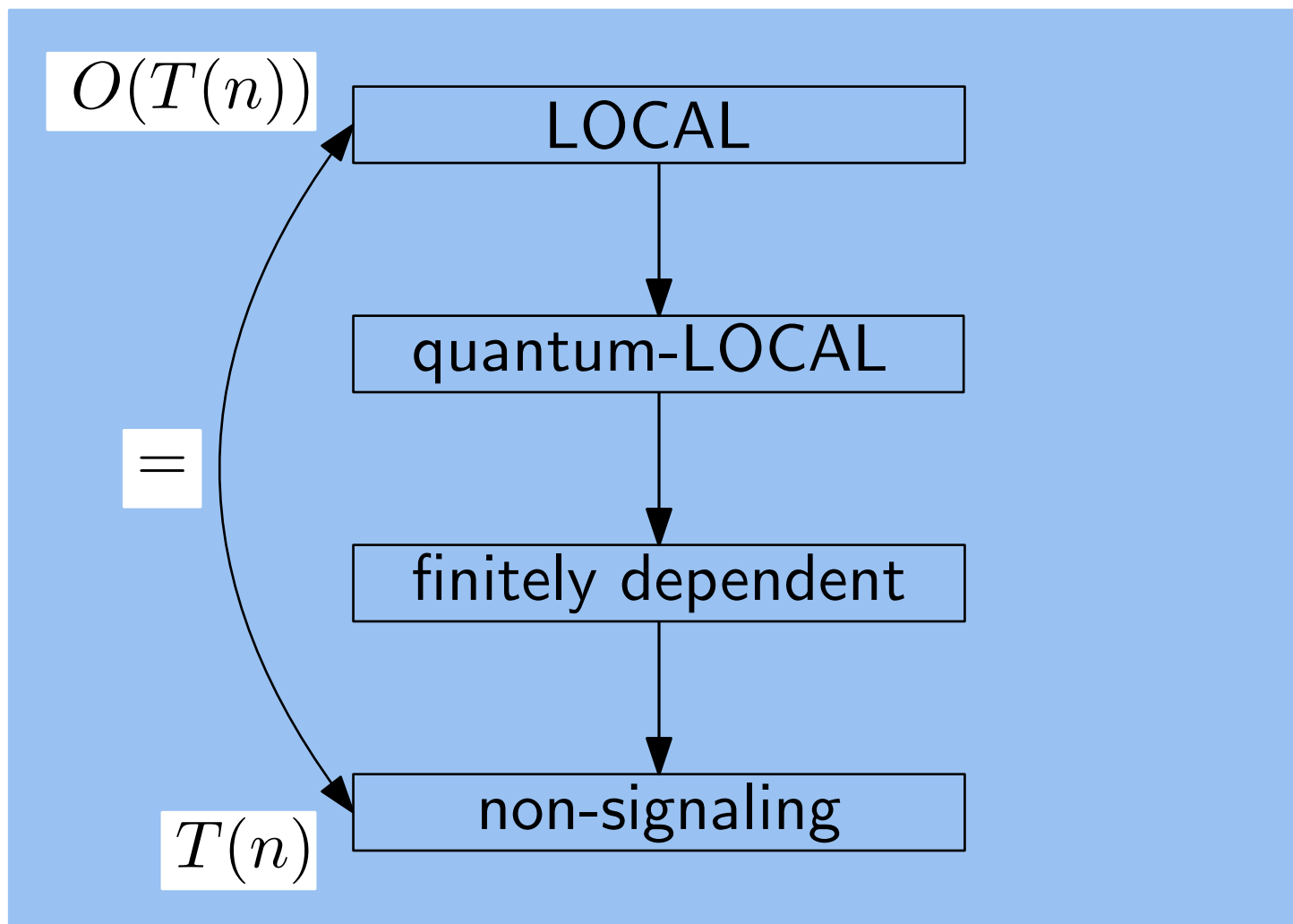


Distributed Hierarchy



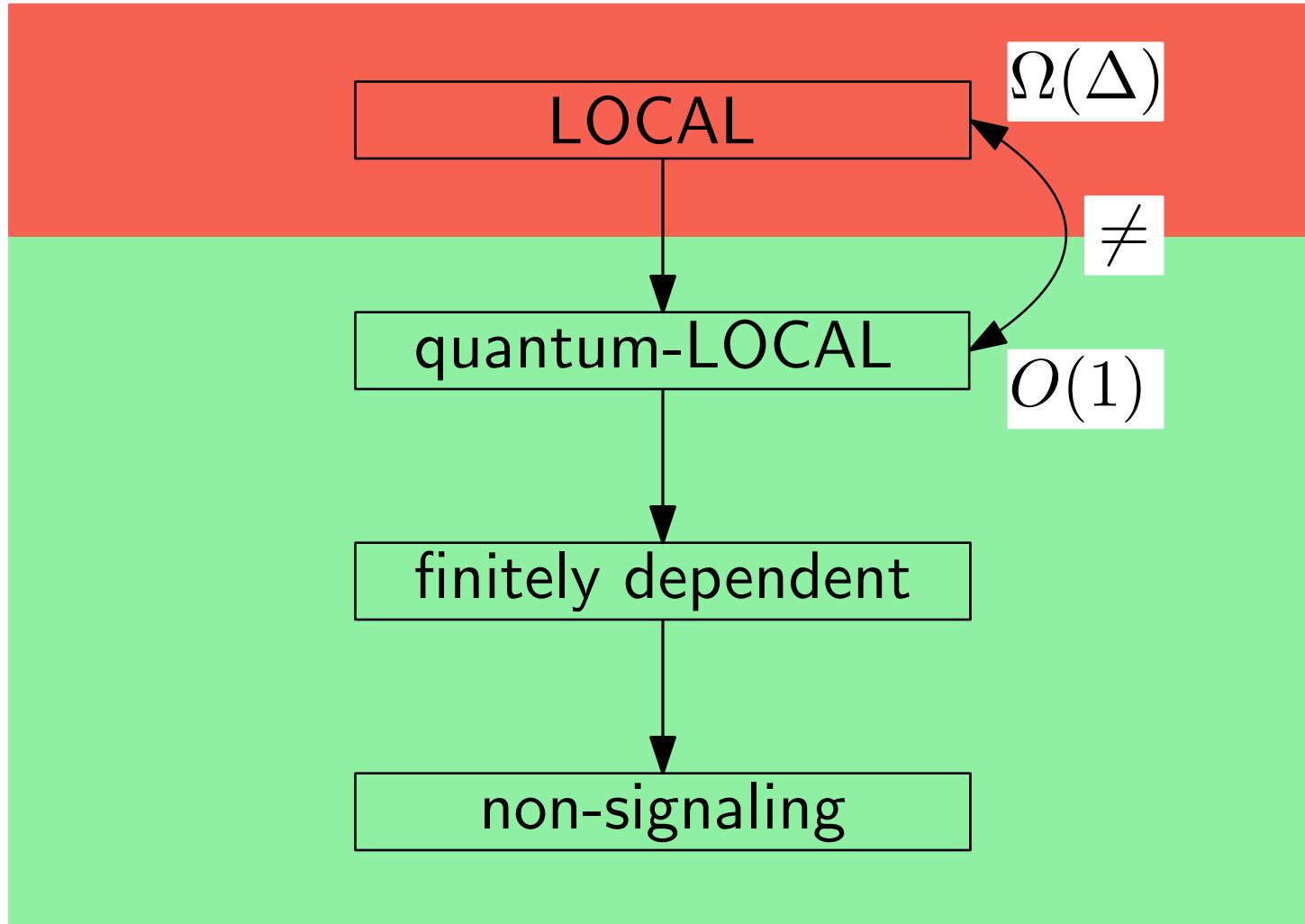
Distributed Hierarchy

For Linear Programs (LPs)

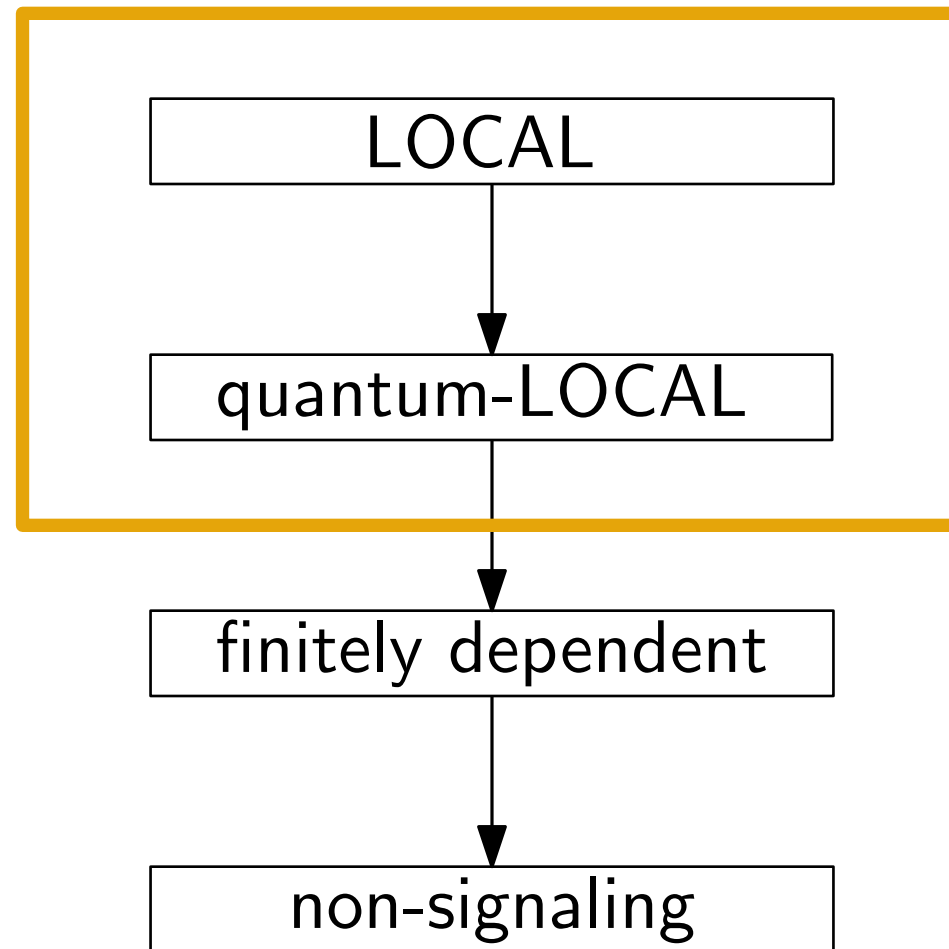


Distributed Hierarchy

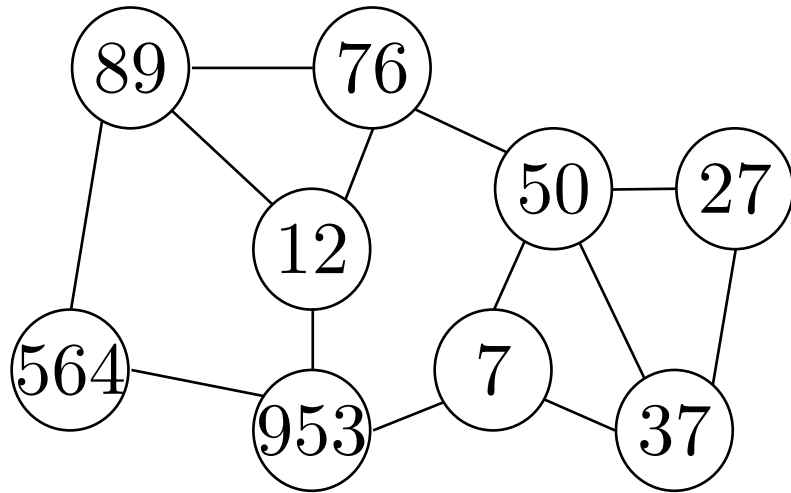
For Locally Checkable Labelling Problems (LCLs)



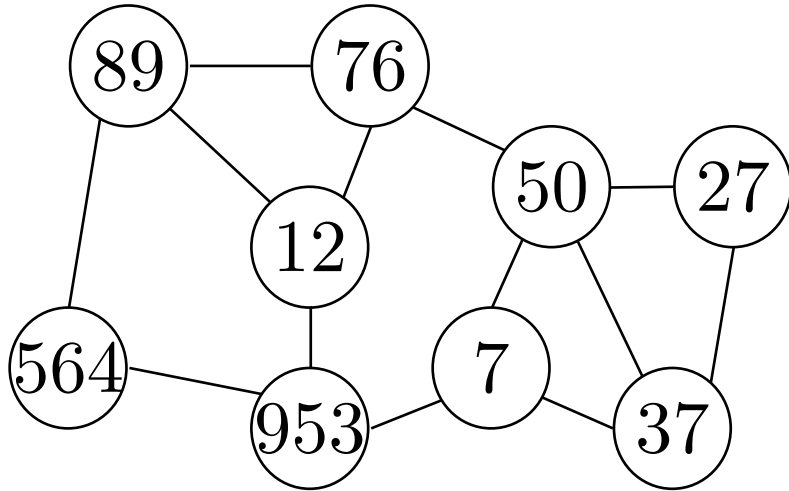
Distributed Hierarchy



LOCAL and quantum-LOCAL

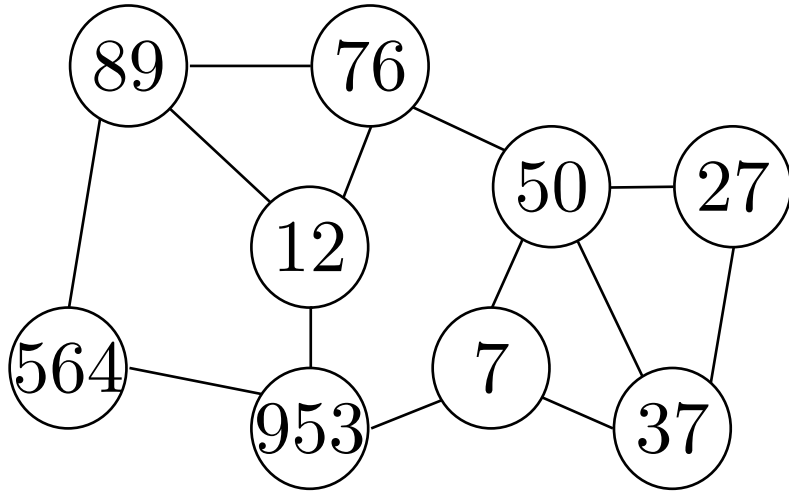


LOCAL and quantum-LOCAL



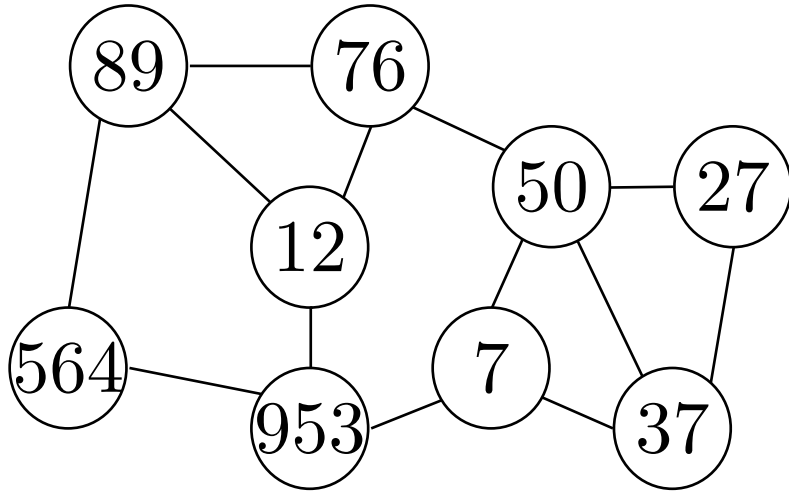
- The network is a graph $G = (V, E)$ of n nodes

LOCAL and quantum-LOCAL



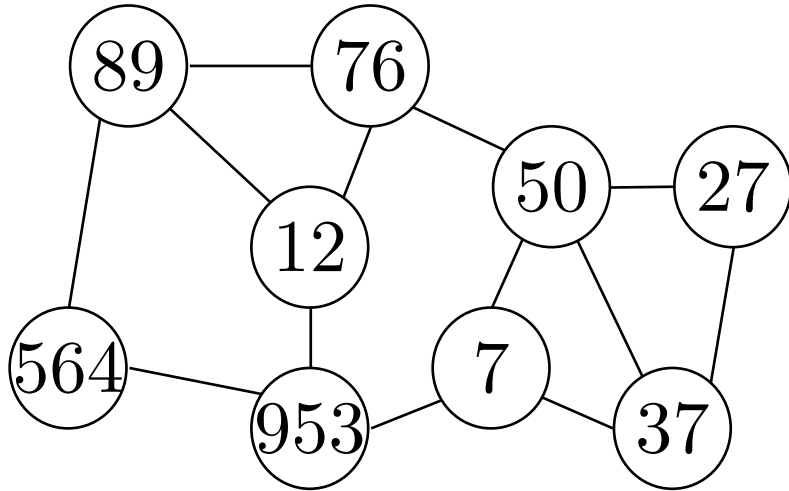
- The network is a graph $G = (V, E)$ of n nodes
- Nodes have input and unique IDs

LOCAL and quantum-LOCAL



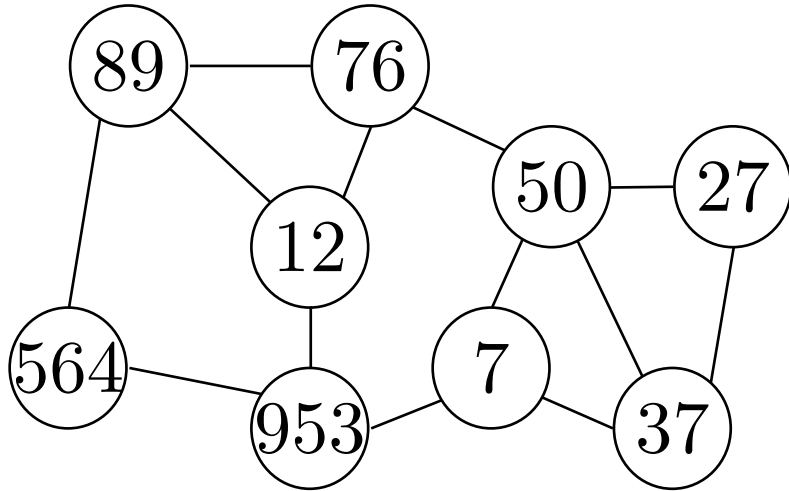
- The network is a graph $G = (V, E)$ of n nodes
- Nodes have input and unique IDs
- IDs belong to $\{1, 2, \dots, n^c\}$

LOCAL and quantum-LOCAL



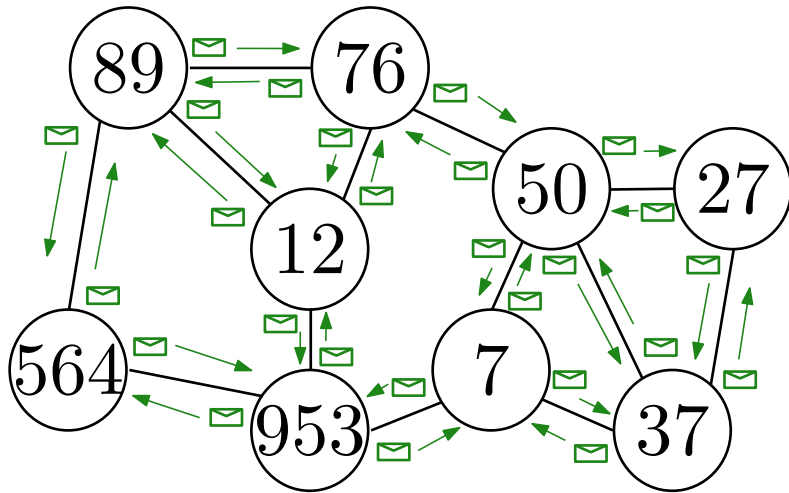
- The network is a graph $G = (V, E)$ of n nodes
- Nodes have input and unique IDs
- IDs belong to $\{1, 2, \dots, n^c\}$
- Computation is **unbounded**

LOCAL and quantum-LOCAL



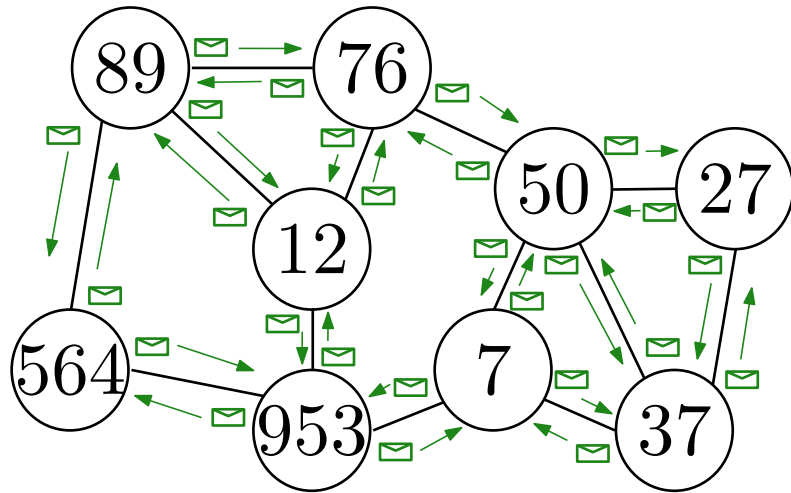
- The network is a graph $G = (V, E)$ of n nodes
- Nodes have input and unique IDs
- IDs belong to $\{1, 2, \dots, n^c\}$
- Computation is **unbounded**
- Message size is **unbounded**

LOCAL and quantum-LOCAL



- The network is a graph $G = (V, E)$ of n nodes
- Nodes have input and unique IDs
- IDs belong to $\{1, 2, \dots, n^c\}$
- Computation is **unbounded**
- Message size is **unbounded**

LOCAL and quantum-LOCAL

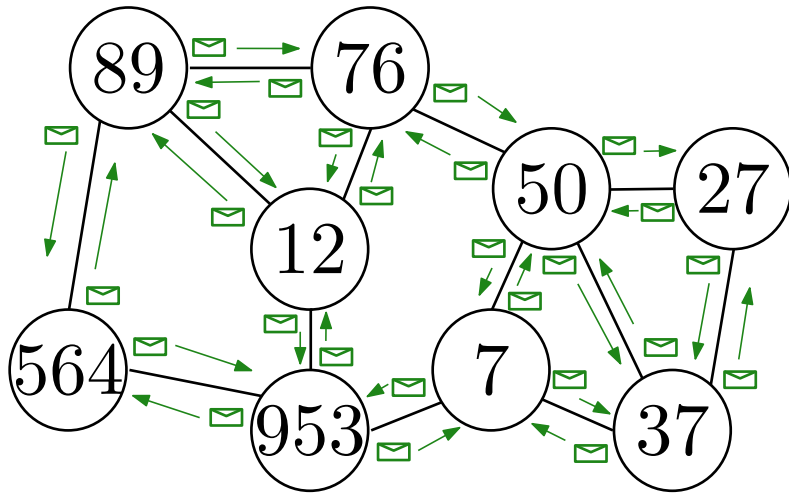


Complexity

Number of communication rounds
as a function of n

- The network is a graph $G = (V, E)$ of n nodes
- Nodes have input and unique IDs
- IDs belong to $\{1, 2, \dots, n^c\}$
- Computation is **unbounded**
- Message size is **unbounded**

LOCAL and quantum-LOCAL



- The network is a graph $G = (V, E)$ of n nodes
- Nodes have input and unique IDs
- IDs belong to $\{1, 2, \dots, n^c\}$
- Computation is **unbounded**
- Message size is **unbounded**

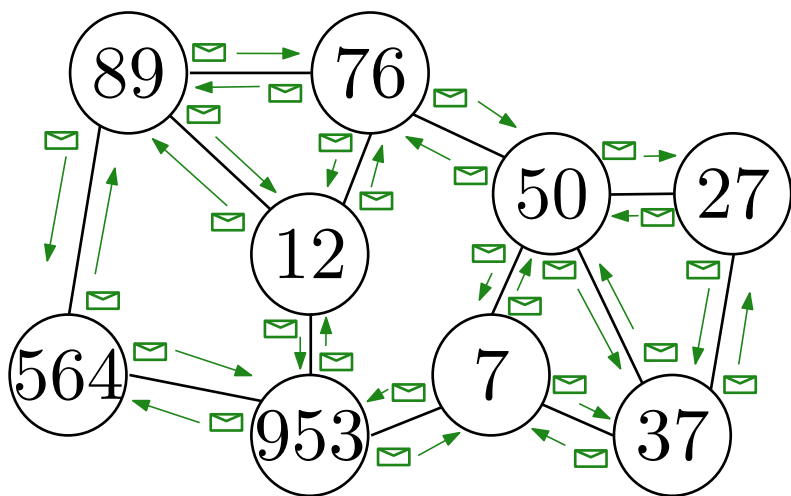
Complexity

Number of communication rounds
as a function of n

LOCAL

Nodes hold, send and receive **bits**

LOCAL and quantum-LOCAL



- The network is a graph $G = (V, E)$ of n nodes
- Nodes have input and unique IDs
- IDs belong to $\{1, 2, \dots, n^c\}$
- Computation is **unbounded**
- Message size is **unbounded**

Complexity

Number of communication rounds
as a function of n

LOCAL

Nodes hold, send and receive **bits**

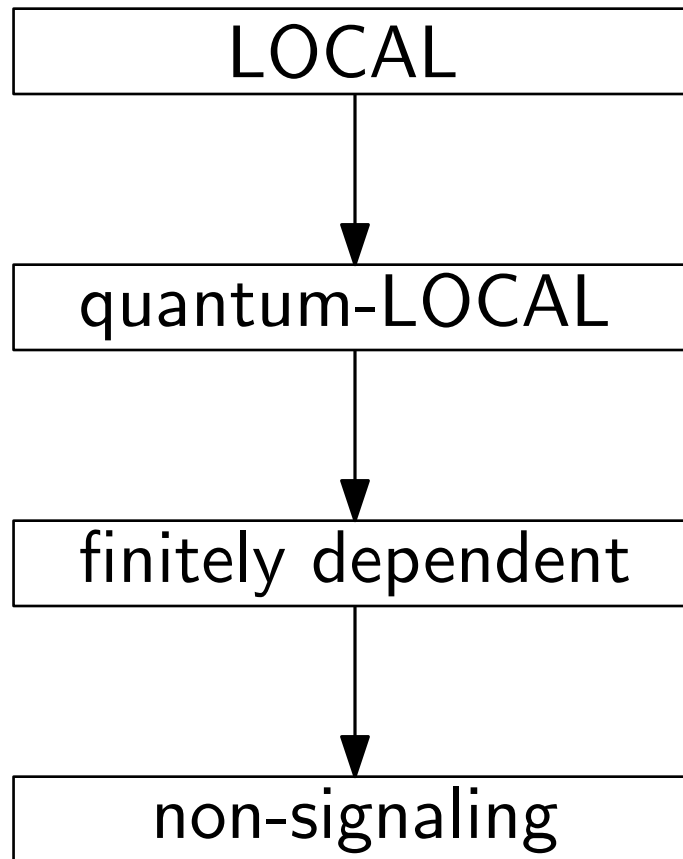
quantum-LOCAL

Nodes hold, send and receive **qubits**

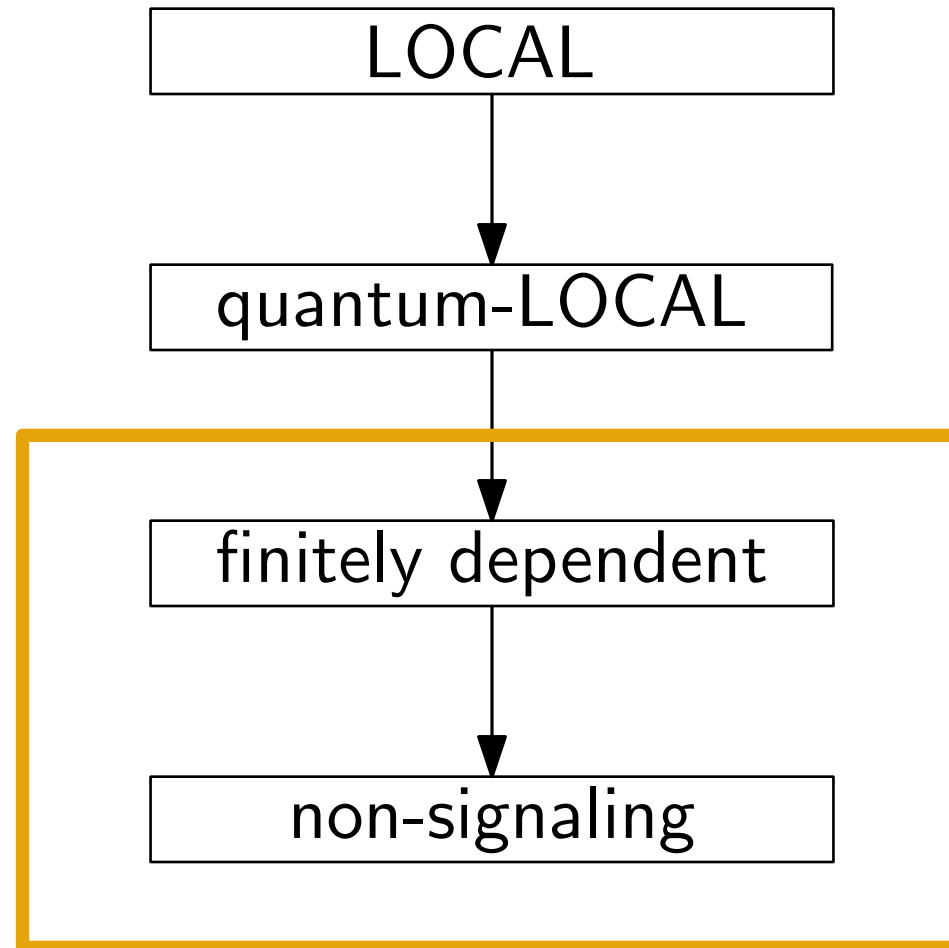
IDs are not needed

“send qubits” means applying
SWAP gates

Distributed Hierarchy

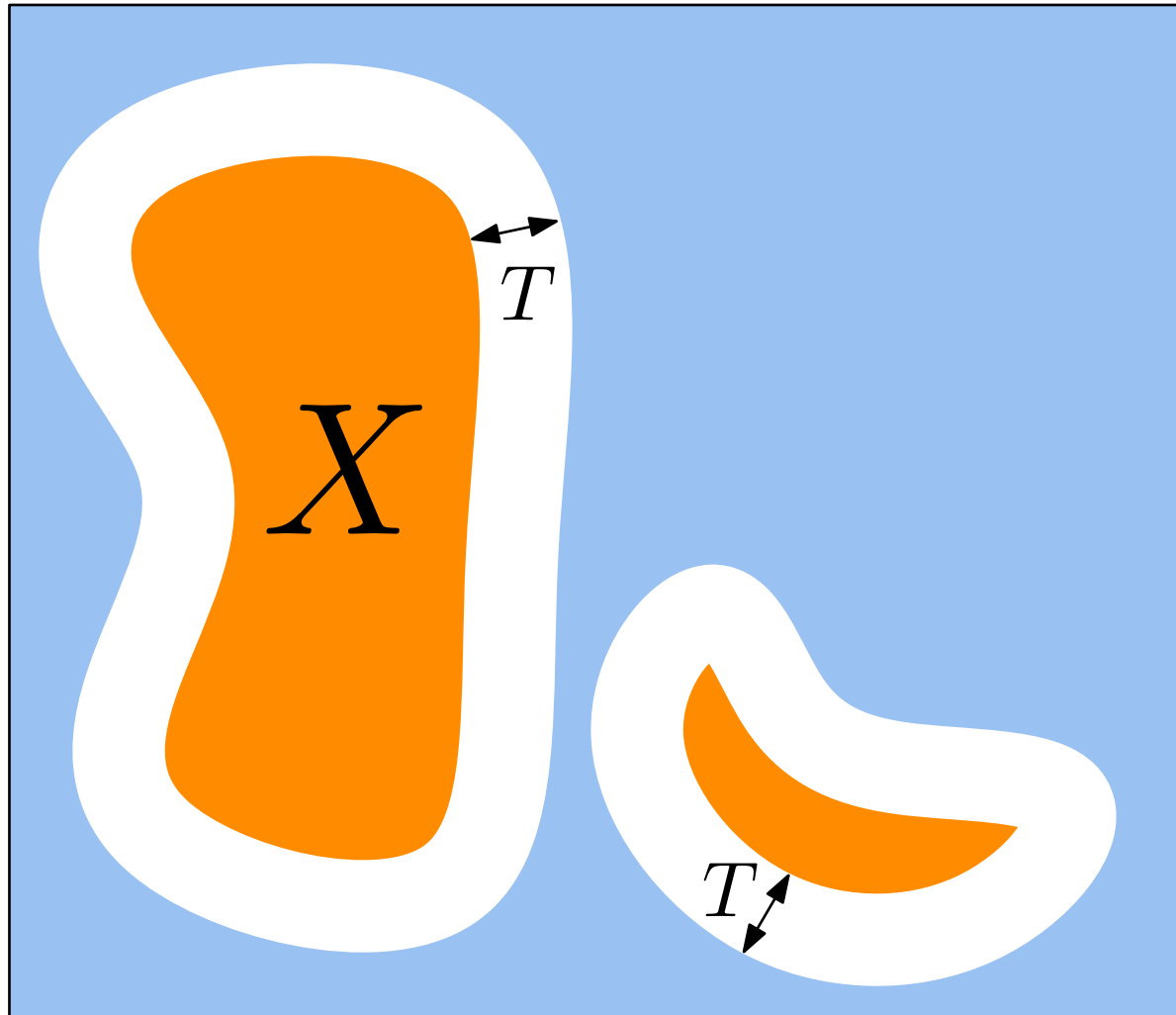


Distributed Hierarchy



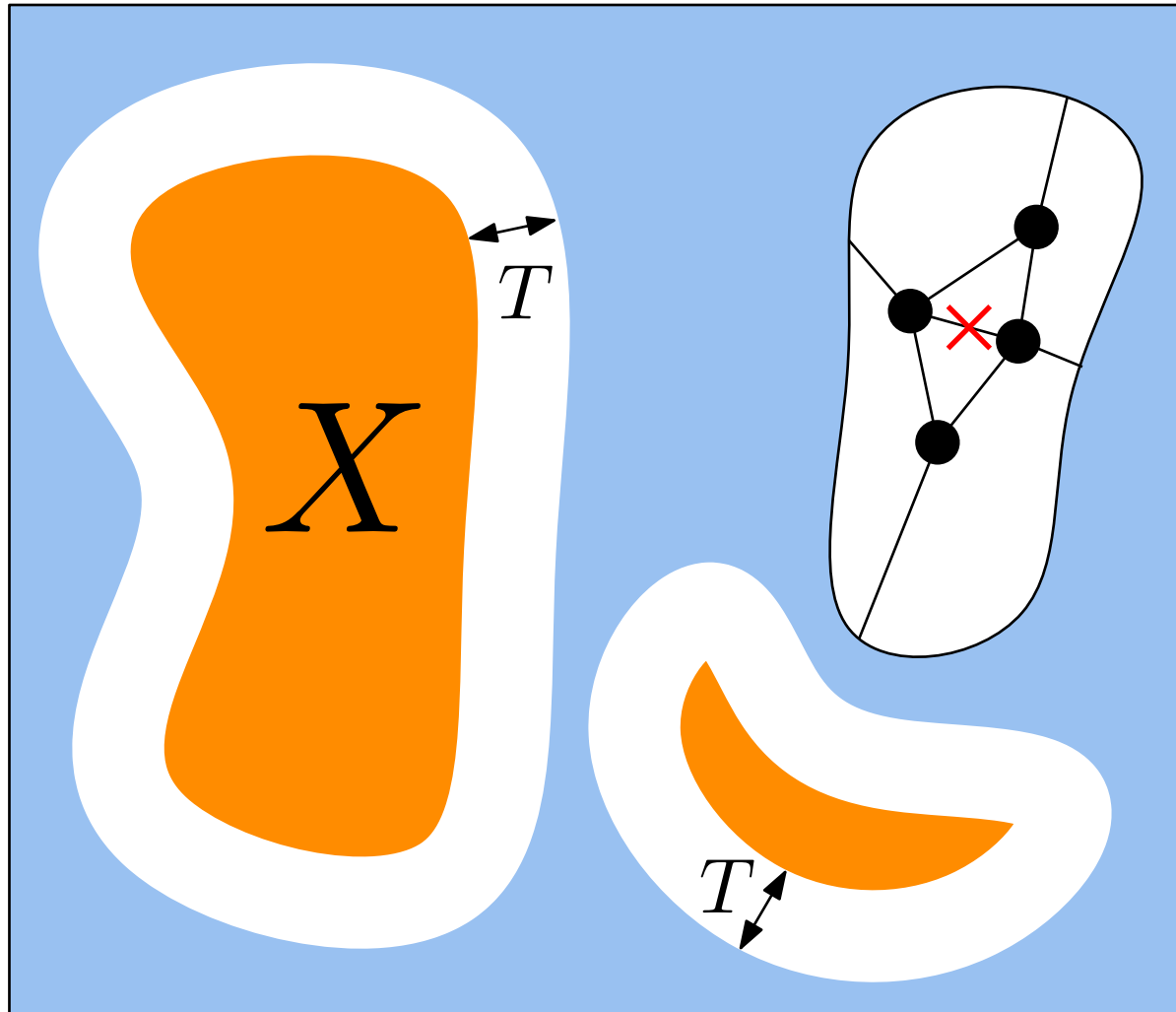
Non-signaling distributions

For any subset of nodes X , changes at distance $> T$
do not affect the output distribution of X



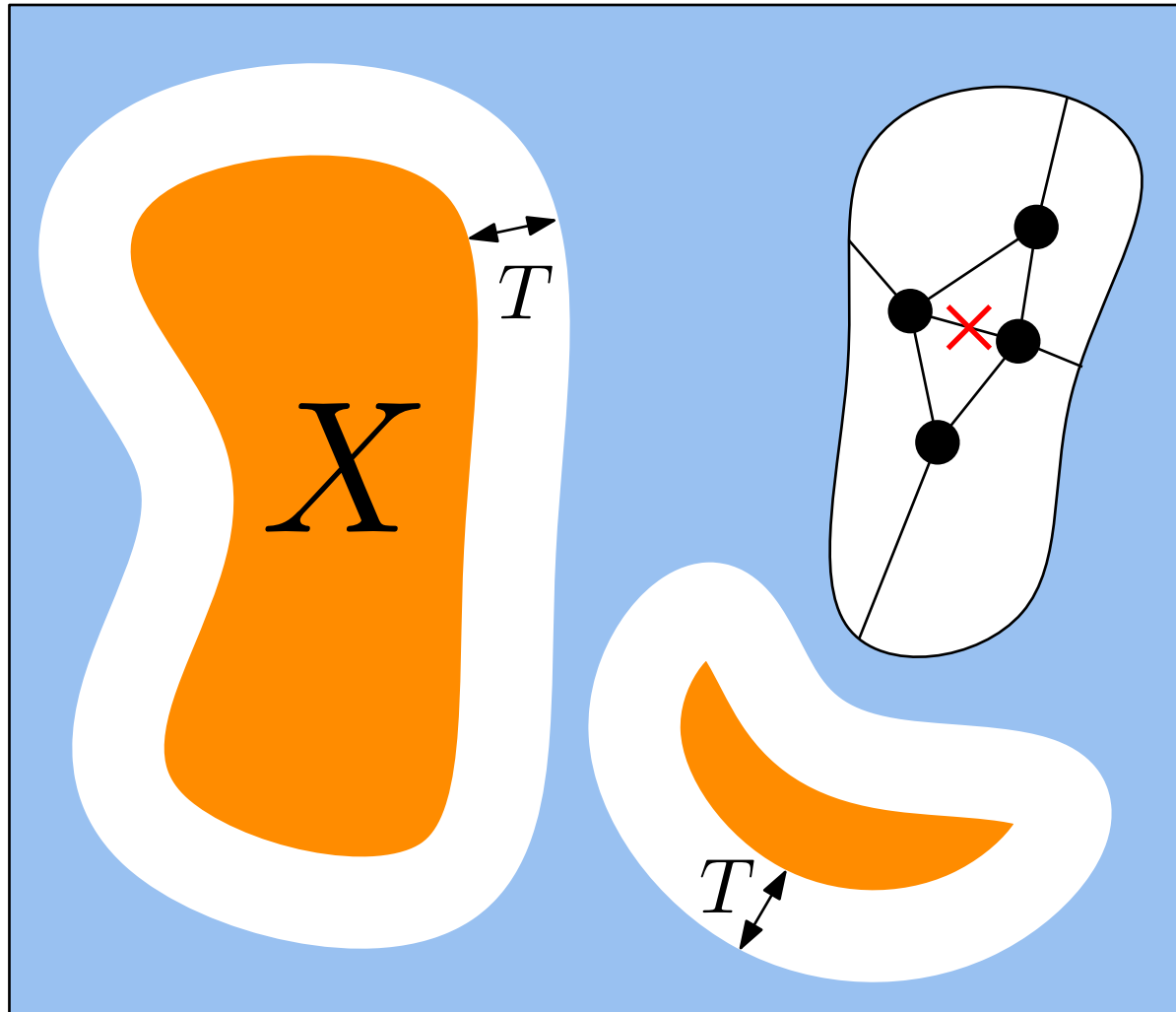
Non-signaling distributions

For any subset of nodes X , changes at distance $> T$
do not affect the output distribution of X



Non-signaling distributions

For any subset of nodes X , changes at distance $> T$
do not affect the output distribution of X



Finetely dependent distributions add restrictions not considered today

Results for Linear Programs (LPs)

Reminder:

- **Independent Set (IS):**

a subset of nodes $I \subseteq V$ such that $\forall v, u \in I : (v, u) \notin E$

- **Maximal Independent Set (MIS):**

a subset of nodes $I \subseteq V$ such that $\forall u \in V \setminus I, \exists v \in I : (v, u) \in E$

Results for Linear Programs (LPs)

Reminder:

- **Independent Set (IS):**

a subset of nodes $I \subseteq V$ such that $\forall v, u \in I : (v, u) \notin E$

- **Maximal Independent Set (MIS):**

a subset of nodes $I \subseteq V$ such that $\forall u \in V \setminus I, \exists v \in I : (v, u) \in E$

Example of Distributed LP: **fractional maximum independent set**

$$\text{maximize } \sum_{v \in V} x_v$$

$$\text{subject to } \sum_{(v,u) \in E} x_v + x_u \leq 1 \quad \forall v \in V$$

$$0 \leq x_v \leq 1 \quad \forall v \in V$$

Results for Linear Programs (LPs)

Reminder:

- **Independent Set (IS):**

a subset of nodes $I \subseteq V$ such that $\forall v, u \in I : (v, u) \notin E$

- **Maximal Independent Set (MIS):**

a subset of nodes $I \subseteq V$ such that $\forall u \in V \setminus I, \exists v \in I : (v, u) \in E$

Example of Distributed LP: **fractional maximum independent set**

$$\text{maximize } \sum_{v \in V} x_v$$

$$\text{subject to } \sum_{(v,u) \in E} x_v + x_u \leq 1 \quad \forall v \in V$$

$$0 \leq x_v \leq 1 \quad \forall v \in V$$

When $x_v \in \{0, 1\} \quad \forall v \in V$, the solution is a **maximum independent set**, i.e. an MIS of maximum cardinality.

Results for Linear Programs (LPs)

For LPs, LOCAL can simulate any non-signaling distribution

Example for fractional MIS

Let $T(n)$ be the locality of the non-sign. distribution

Results for Linear Programs (LPs)

For LPs, LOCAL can simulate any non-signaling distribution

Example for fractional MIS

Let $T(n)$ be the locality of the non-sign. distribution

LOCAL algorithm:

- As node v , explore your radius- $T(n)$ neighborhood
- Set the output for v to be $\mathbb{E}[X_v]$

Results for Linear Programs (LPs)

For LPs, LOCAL can simulate any non-signaling distribution

Example for fractional MIS

Let $T(n)$ be the locality of the non-sign. distribution

LOCAL algorithm:

- As node v , explore your radius- $T(n)$ neighborhood
- Set the output for v to be $\mathbb{E}[X_v]$

Here X_v is the rand. var. for the output of node v

Results for Linear Programs (LPs)

For LPs, LOCAL can simulate any non-signaling distribution

Example for fractional MIS

Let $T(n)$ be the locality of the non-sign. distribution

LOCAL algorithm:

- As node v , explore your radius- $T(n)$ neighborhood
- Set the output for v to be $\mathbb{E}[X_v]$

Here X_v is the rand. var. for the output of node v

Let $X = \sum_{v \in V} X_v$ be the rand. var. for the output of the entire graph

Results for Linear Programs (LPs)

For LPs, LOCAL can simulate any non-signaling distribution

Example for fractional MIS

Let $T(n)$ be the locality of the non-sign. distribution

LOCAL algorithm:

- As node v , explore your radius- $T(n)$ neighborhood
- Set the output for v to be $\mathbb{E}[X_v]$

Here X_v is the rand. var. for the output of node v

Let $X = \sum_{v \in V} X_v$ be the rand. var. for the output of the entire graph

Key facts

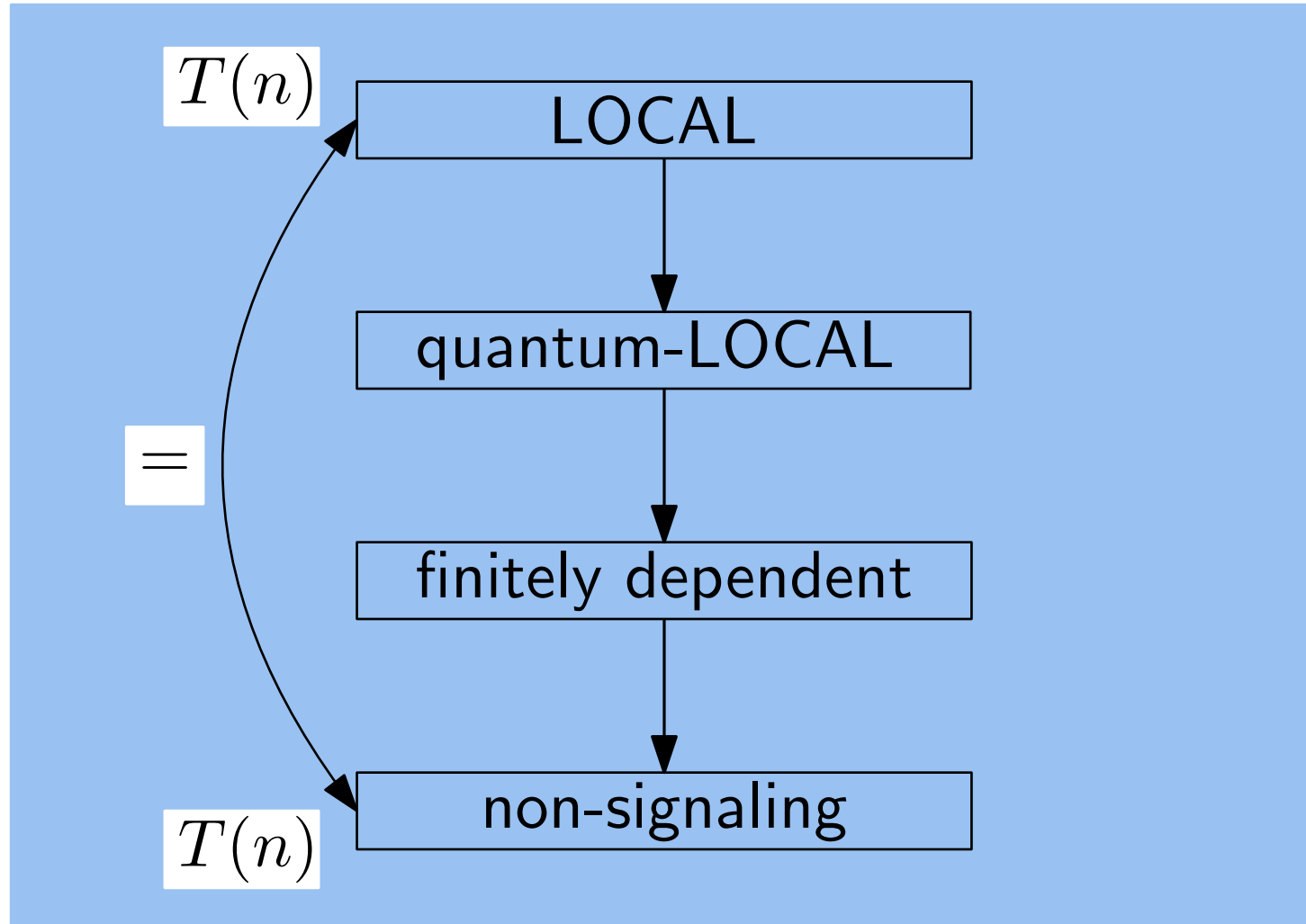
- By linearity of $\mathbb{E}$,

$$\sum_{v \in V} \mathbb{E}[X_v] = \mathbb{E} \left[\sum_{v \in V} X_v \right] = \mathbb{E}[X]$$

- By monotonicity of $\mathbb{E}$, if X is an α -approx. then $\mathbb{E}[X]$ is an α -approx.

Results for Linear Programs (LPs)

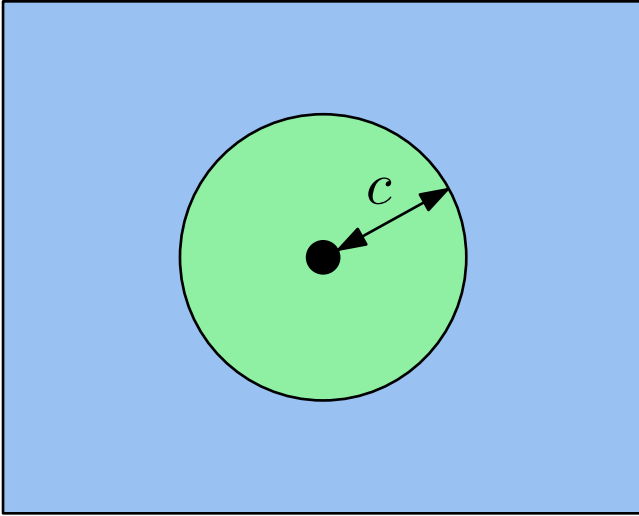
For Linear Programs (LPs)



Results for Locally Checable Labelly Problembs (LCLs)

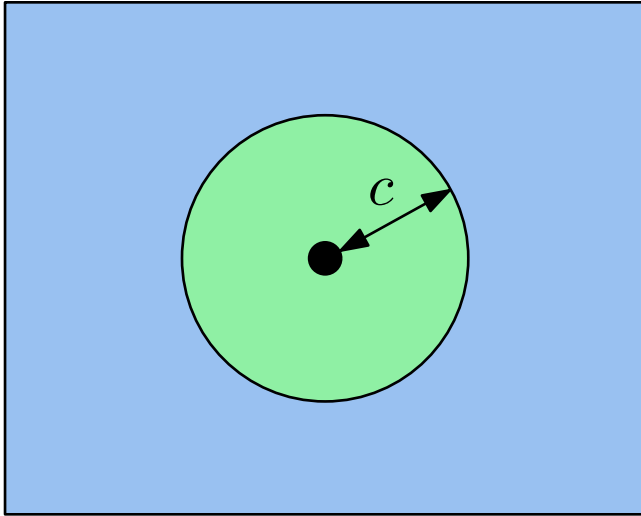
Results for Locally Checable Labelly Problembs (LCLs)

LCLs: explore up to radius c to check a solution



Results for Locally Checable Labelly Problembs (LCLs)

LCLs: explore up to radius c to check a solution

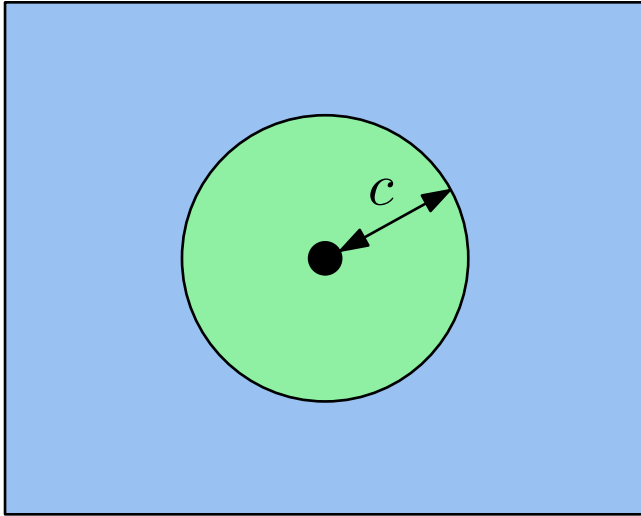


Examples

- MIS, maximal instead of (fractional) maximum
- Coloring
- Maximal Matching
- iterated-GHZ

Results for Locally Checable Labelly Problems (LCLs)

LCLs: explore up to radius c to check a solution



Examples

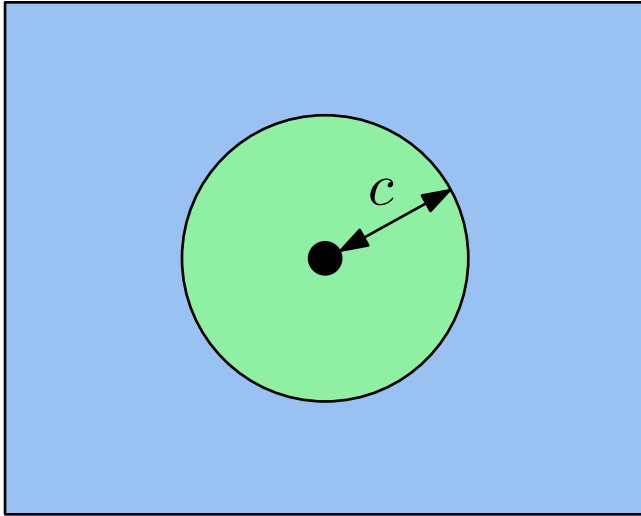
- MIS, maximal instead of (fractional) maximum
- Coloring
- Maximal Matching
- iterated-GHZ

Reminder on GHZ

Three players A , B and C , and one referee

Results for Locally Checable Labelly Problembs (LCLs)

LCLs: explore up to radius c to check a solution



Examples

- MIS, maximal instead of (fractional) maximum
- Coloring
- Maximal Matching
- iterated-GHZ

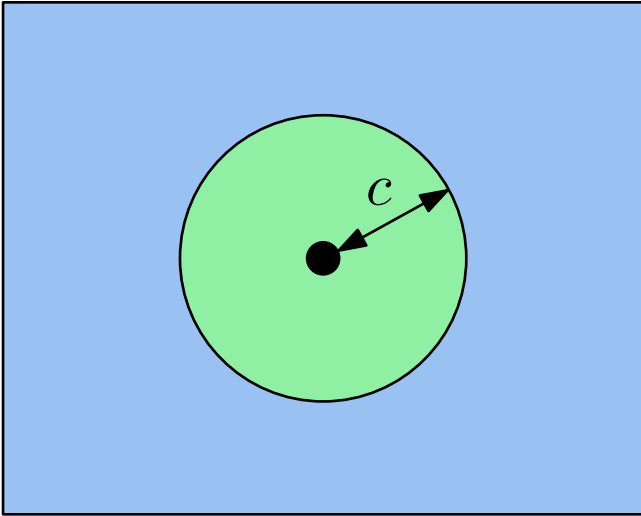
Reminder on GHZ

Three players A , B and C , and one referee

inputs x , y and z ; outputs: a , b and c

Results for Locally Checable Labelly Problems (LCLs)

LCLs: explore up to radius c to check a solution



Examples

- MIS, maximal instead of (fractional) maximum
- Coloring
- Maximal Matching
- iterated-GHZ

Reminder on GHZ

Three players A , B and C , and one referee

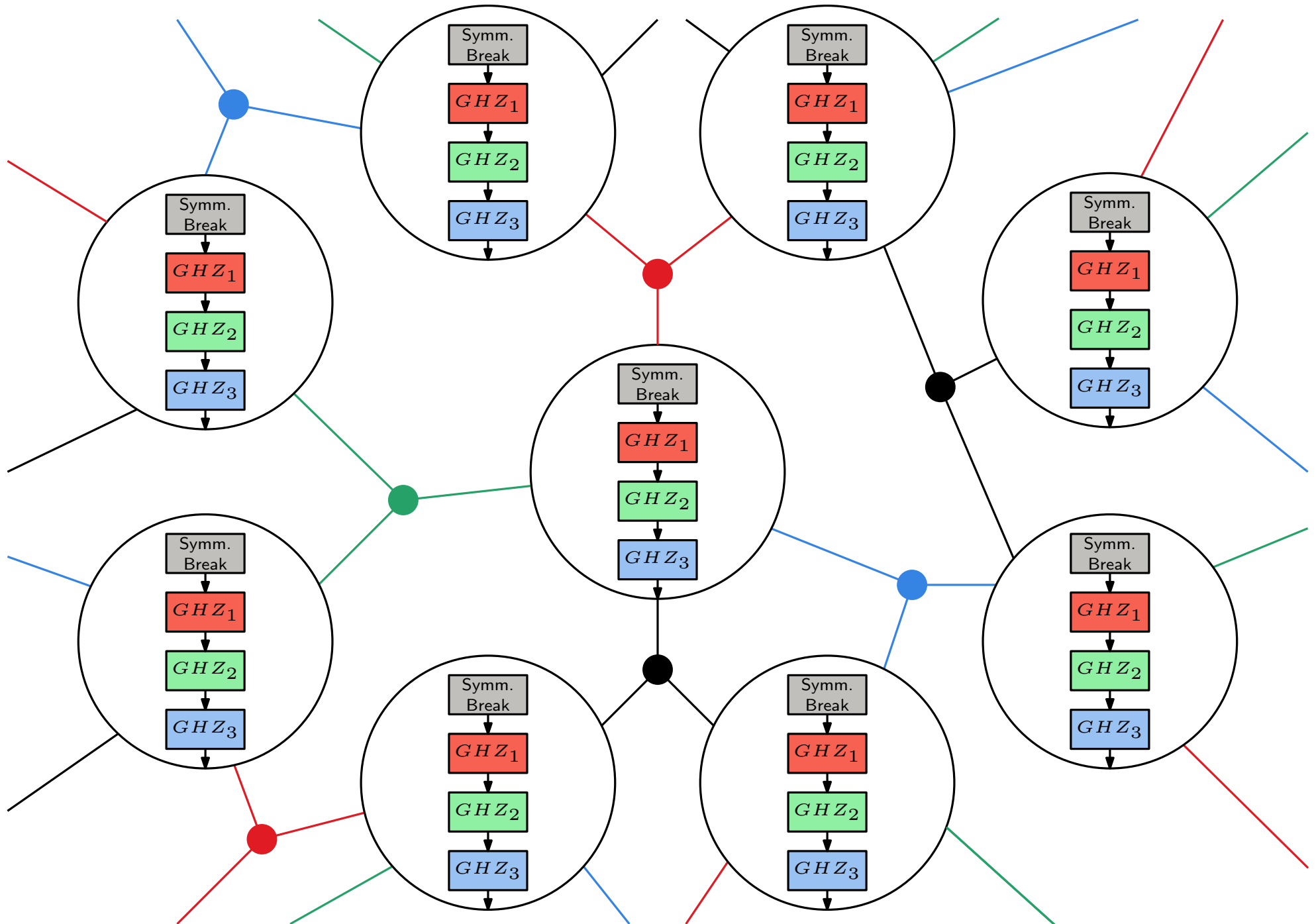
inputs x , y and z ; outputs: a , b and c

Goal: **without communicating after the input is given**, output bits such that

$$a \oplus b \oplus c = x \vee y \vee z$$

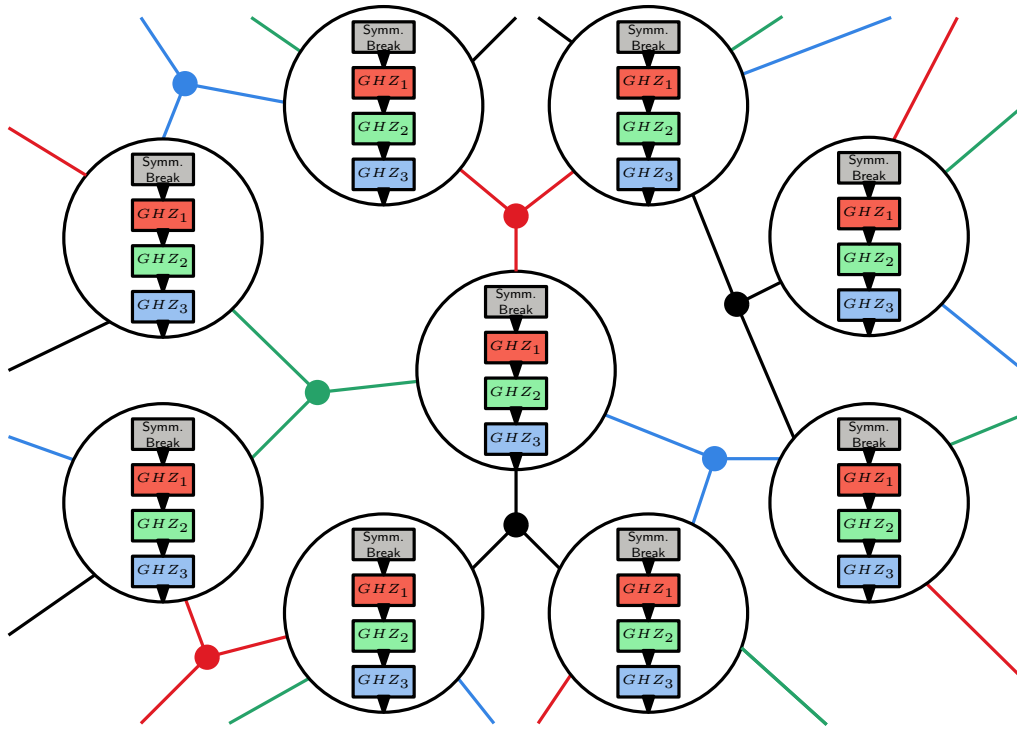
knowing that $x \vee y \vee z \in \{(0, 0, 0), (1, 1, 0), (1, 0, 1), (0, 1, 1)\}$

Results for Locally Checable Labelly Problembs (LCLs)



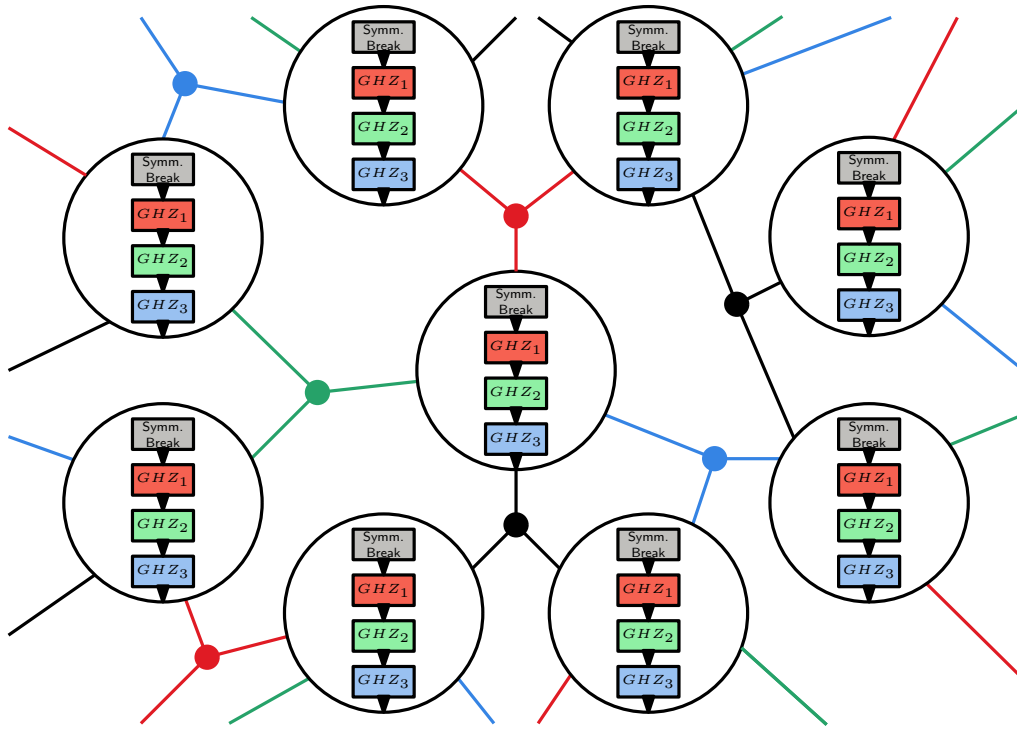
Results for Locally Checable Labelly Problembs (LCLs)

Round complexity of iterated-GHZ



Results for Locally Checable Labelly Problembs (LCLs)

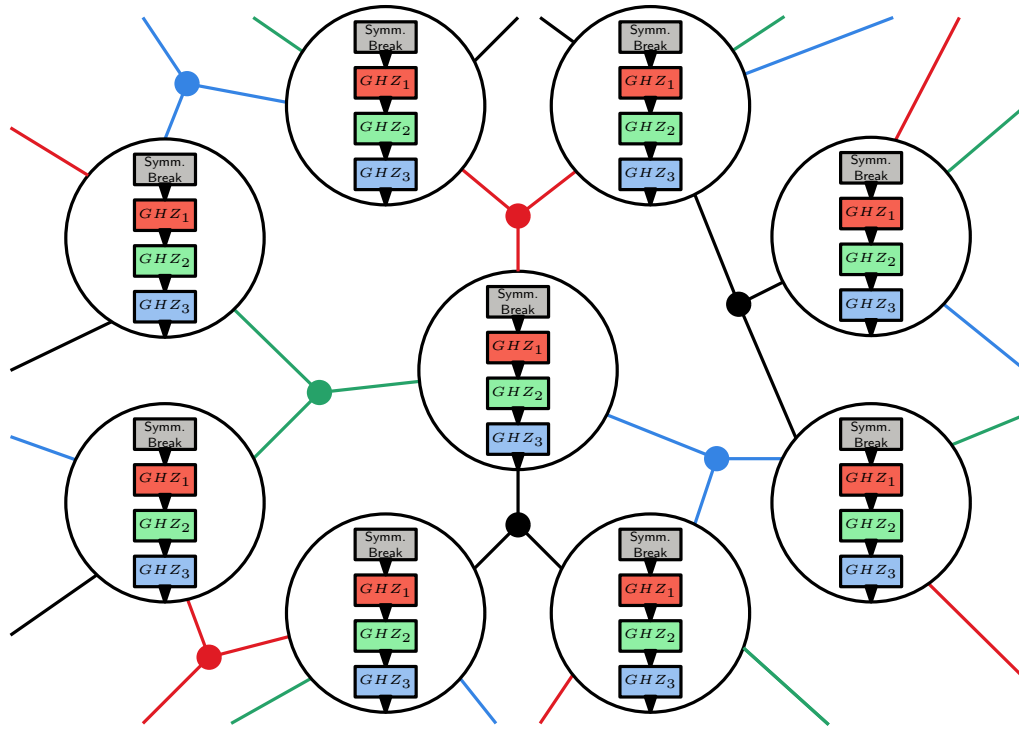
Round complexity of iterated-GHZ



quantum-LOCAL: $O(1)$ rounds

Results for Locally Checable Labelly Problembs (LCLs)

Round complexity of iterated-GHZ

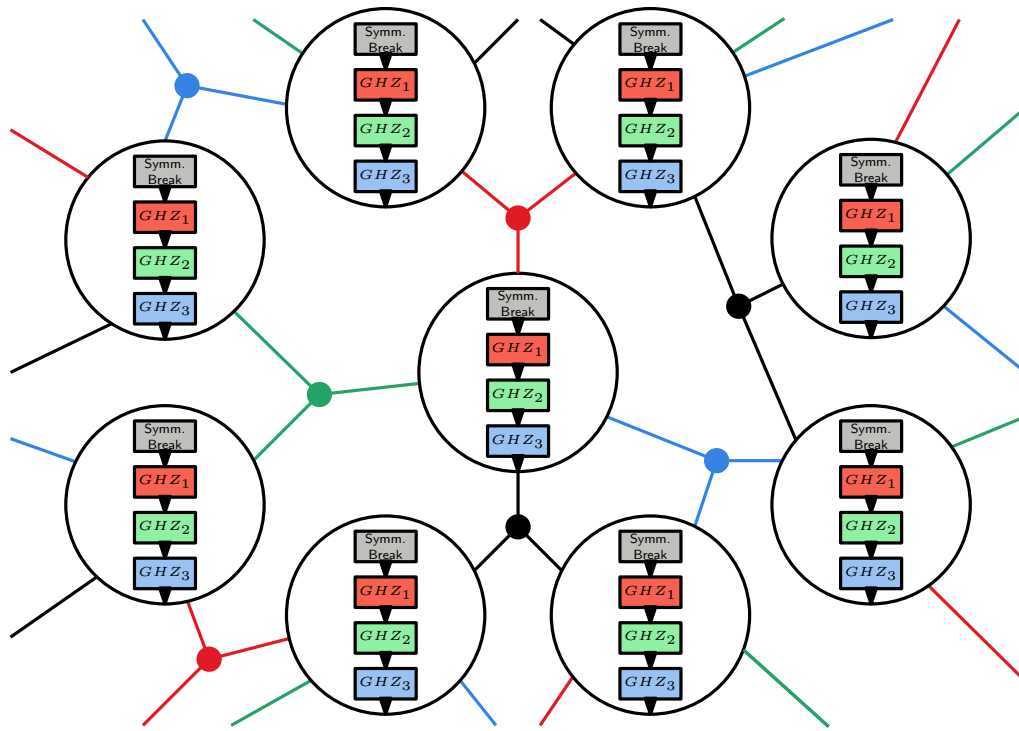


quantum-LOCAL: $O(1)$ rounds

- share a GHZ state

Results for Locally Checable Labelly Problembs (LCLs)

Round complexity of iterated-GHZ

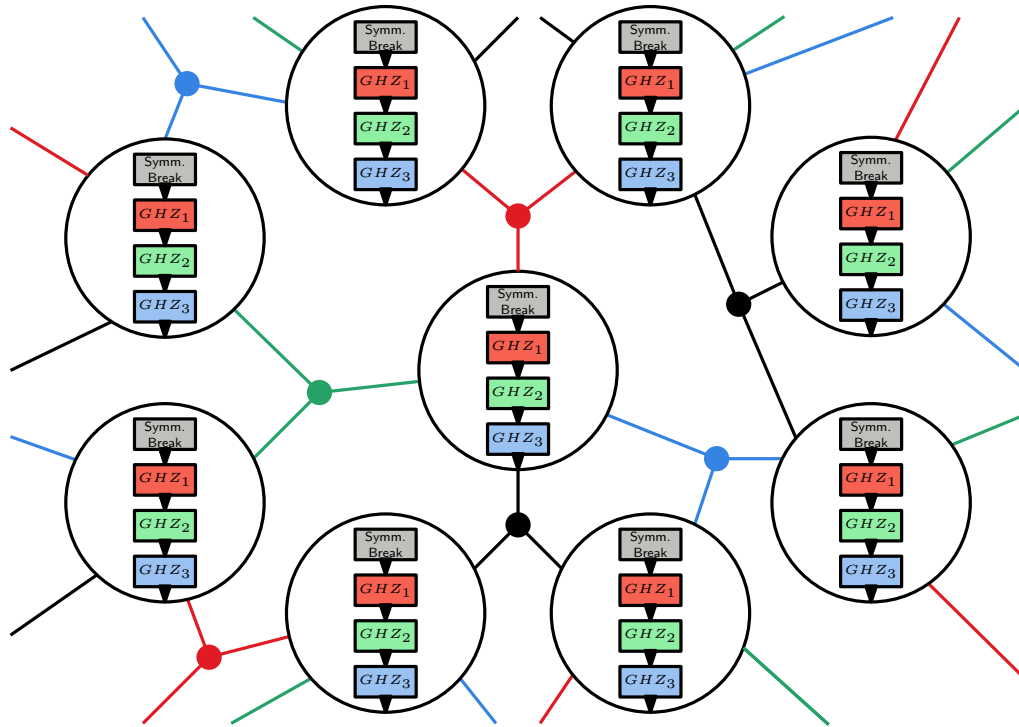


quantum-LOCAL: $O(1)$ rounds

- share a GHZ state
- measure

Results for Locally Checable Labelly Problembs (LCLs)

Round complexity of iterated-GHZ



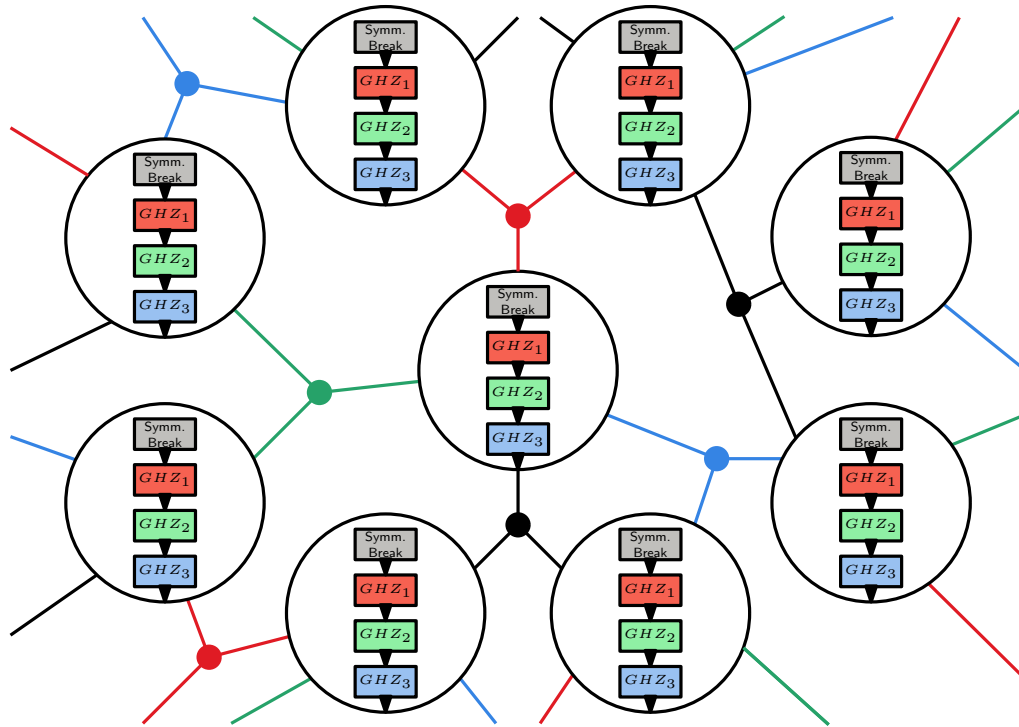
quantum-LOCAL: $O(1)$ rounds

- share a GHZ state
- measure

LOCAL: $O(\Delta)$ rounds

Results for Locally Checable Labelly Problembs (LCLs)

Round complexity of iterated-GHZ



quantum-LOCAL: $O(1)$ rounds

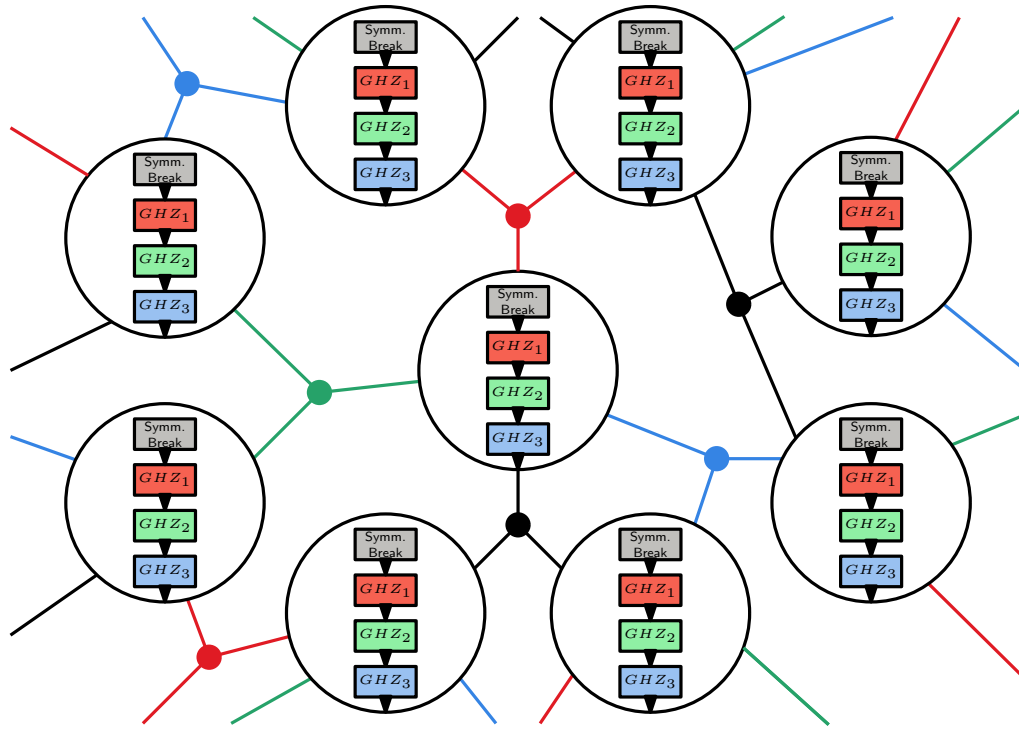
- share a GHZ state
- measure

LOCAL: $O(\Delta)$ rounds

- solve symm. break. with 1st neigh.

Results for Locally Checable Labelly Problembs (LCLs)

Round complexity of iterated-GHZ



quantum-LOCAL: $O(1)$ rounds

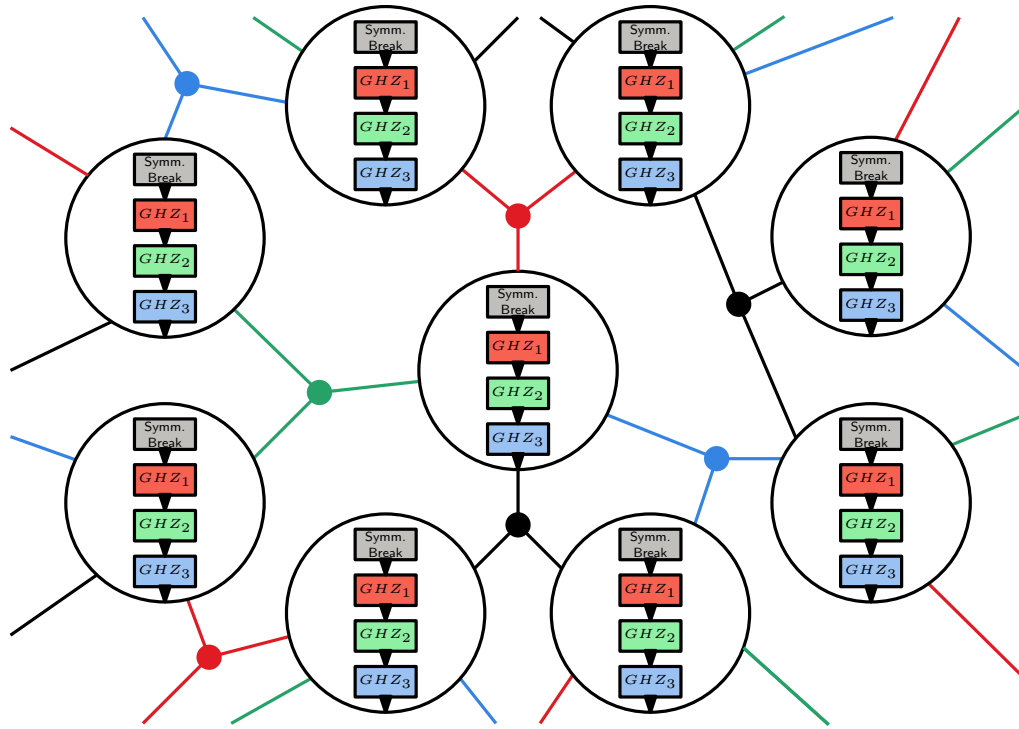
- share a GHZ state
- measure

LOCAL: $O(\Delta)$ rounds

- solve symm. break. with 1st neigh.
- solve GHZ_1 game with 2nd neigh.

Results for Locally Checable Labelly Problembs (LCLs)

Round complexity of iterated-GHZ



quantum-LOCAL: $O(1)$ rounds

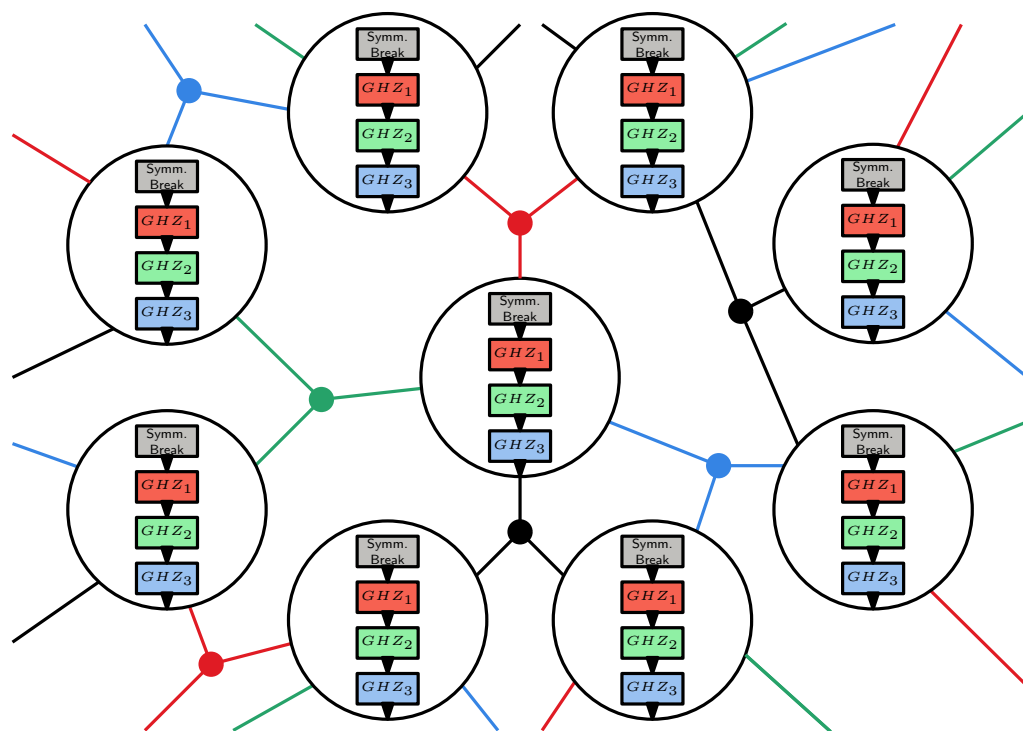
- share a GHZ state
- measure

LOCAL: $O(\Delta)$ rounds

- solve symm. break. with 1st neigh.
- solve GHZ_1 game with 2nd neigh.
- solve GHZ_2 game with 3rd neigh.

Results for Locally Checable Labelly Problembs (LCLs)

Round complexity of iterated-GHZ



quantum-LOCAL: $O(1)$ rounds

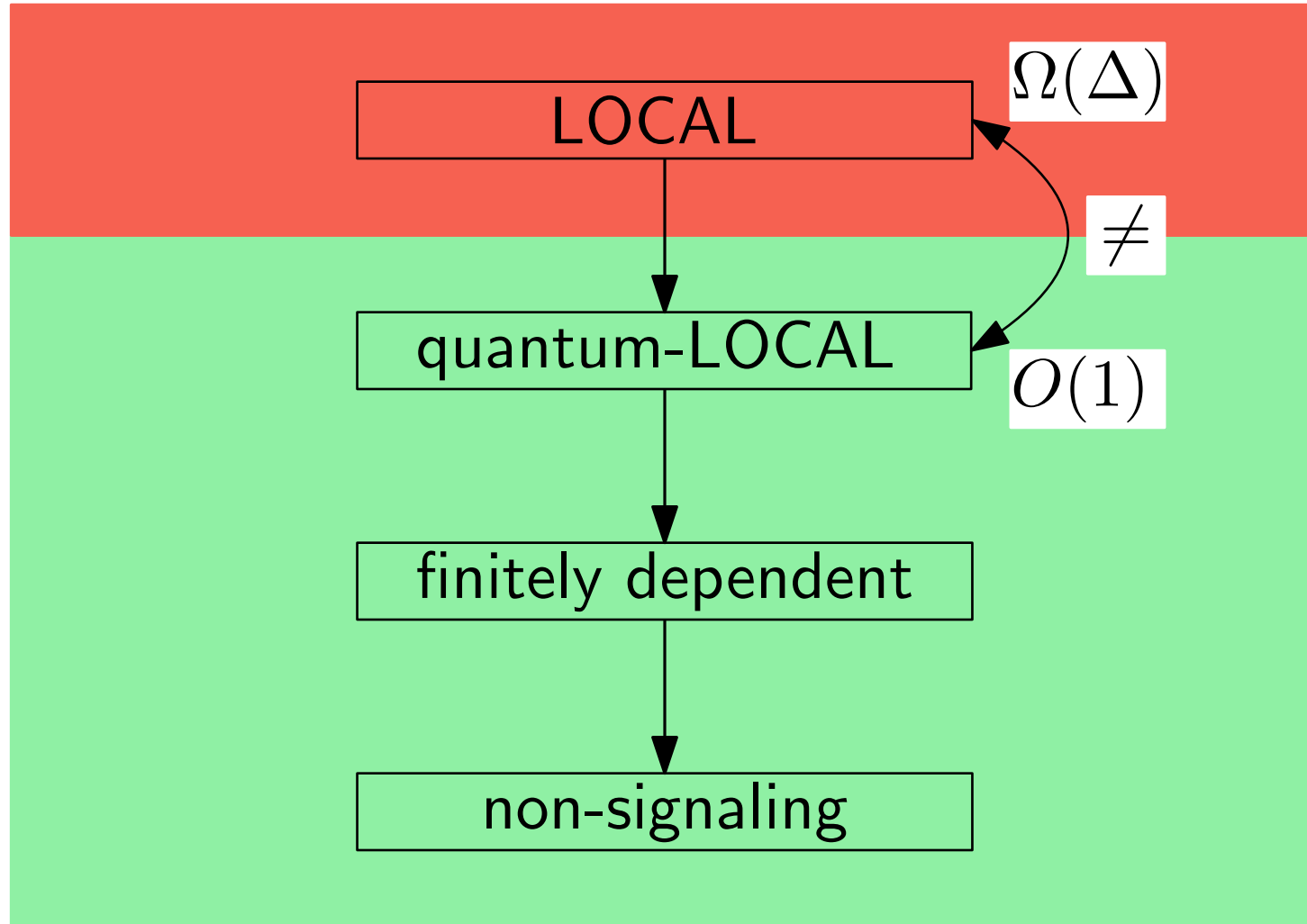
- share a GHZ state
- measure

LOCAL: $O(\Delta)$ rounds

- solve symm. break. with 1st neigh.
- solve GHZ_1 game with 2nd neigh.
- solve GHZ_2 game with 3rd neigh.
- ...
- solve GHZ_Δ game with the last neigh.

Results for Locally Checkable Labelling Problems (LCLs)

For Locally Checkable Labelling Problems (LCLs)



Summary

Summary

Linear Programs (also SDPs) : no separation

New Limits on Distributed Quantum Advantage: Dequantizing Linear Programs

A. Balliu, C. Coupet, A. Cruciani, F. D'Amore, M. Equi, H. Lievonen, A. Modanese, D. Olivetti, J. Suomela

DISC 2025

Summary

Linear Programs (also SDPs) : no separation

New Limits on Distributed Quantum Advantage: Dequantizing Linear Programs

A. Balliu, C. Coupette, A. Cruciani, F. D'Amore, M. Equi, H. Lievonen, A. Modanese, D. Olivetti, J. Suomela

DISC 2025

- previously assumed to be true but never formalized
- allows us to prove stronger results

Summary

Linear Programs (also SDPs) : no separation

New Limits on Distributed Quantum Advantage: Dequantizing Linear Programs

A. Balliu, C. Coupette, A. Cruciani, F. D'Amore, M. Equi, H. Lievonen, A. Modanese, D. Olivetti, J. Suomela

DISC 2025

- previously assumed to be true but never formalized
- allows us to prove stronger results

LCLs : $O(\Delta)$ LOCAL vs $O(1)$ quantum-LOCAL

Distributed Quantum Advantage for Local Problems

A. Balliu, S. Brandt, X. Coiteux-Roy, F. d'Amore, M. Equi, F. Le Gall, H. Lievonen, A. Modanese, D. Olivetti, M. Renou, J. Suomela, L. Tendick, I. Veeren

STOC 2025

... later improved in SODA 2026

Summary

Linear Programs (also SDPs) : no separation

New Limits on Distributed Quantum Advantage: Dequantizing Linear Programs

A. Balliu, C. Coupette, A. Cruciani, F. D'Amore, M. Equi, H. Lievonen, A. Modanese, D. Olivetti, J. Suomela

DISC 2025

- previously assumed to be true but never formalized
- allows us to prove stronger results

LCLs : $O(\Delta)$ LOCAL vs $O(1)$ quantum-LOCAL

Distributed Quantum Advantage for Local Problems

A. Balliu, S. Brandt, X. Coiteux-Roy, F. d'Amore, M. Equi, F. Le Gall, H. Lievonen, A. Modanese, D. Olivetti, M. Renou, J. Suomela, L. Tendick, I. Veeren

STOC 2025

... later improved in SODA 2026

- only one previous result for LOCAL, and not for LCLs
- most previous literature focused on other models like CONGEST

Summary

Summary

Distributed Quantum Computing

New Limits on Distributed Quantum Advantage: Dequantizing Linear Programs

A. Balliu, C. Coupette, A. Cruciani, F. D'Amore, M. Equi, H. Lievonen, A. Modanese, D. Olivetti, J. Suomela

DISC 2025

Distributed Quantum Advantage for Local Problems

A. Balliu, S. Brandt, X. Coiteux-Roy, F. d'Amore, M. Equi, F. Le Gall, H. Lievonen, A. Modanese, D. Olivetti, M. Renou, J. Suomela, L. Tendick, I. Veeren

STOC 2025

... later improved in SODA 2026

Summary

Distributed Quantum Computing

New Limits on Distributed Quantum Advantage: Dequantizing Linear Programs

A. Balliu, C. Coupette, A. Cruciani, F. D'Amore, M. Equi, H. Lievonen, A. Modanese, D. Olivetti, J. Suomela

DISC 2025

Distributed Quantum Advantage for Local Problems

A. Balliu, S. Brandt, X. Coiteux-Roy, F. d'Amore, M. Equi, F. Le Gall, H. Lievonen, A. Modanese, D. Olivetti, M. Renou, J. Suomela, L. Tendick, I. Veeren

STOC 2025

... later improved in SODA 2026

Quantum String Matching

From Bit-Parallelism to Quantum String Matching for Labelled Graphs

M. Equi, H. Lievonen, A. Meijer-van de Griend, V. Mäkinen

CPM 2023

Quantum Pattern Matching in Generalised Degenerate Strings

M. Equi, H. Lievonen, Md R. Islam Khan, V. Mäkinen

arXiv 2026, under submission

Summary

Distributed Quantum Computing

New Limits on Distributed Quantum Advantage: Dequantizing Linear Programs

A. Balliu, C. Coupette, A. Cruciani, F. D'Amore, M. Equi, H. Lievonen, A. Modanese, D. Olivetti, J. Suomela

DISC 2025

Distributed Quantum Advantage for Local Problems

A. Balliu, S. Brandt, X. Coiteux-Roy, F. d'Amore, M. Equi, F. Le Gall, H. Lievonen, A. Modanese, D. Olivetti, M. Renou, J. Suomela, L. Tendick, I. Veeren

STOC 2025

... later improved in SODA 2026

Quantum String Matching

From Bit-Parallelism to Quantum String Matching for Labelled Graphs

M. Equi, H. Lievonen, A. Meijer-van de Griend, V. Mäkinen

CPM 2023

Quantum Pattern Matching in Generalised Degenerate Strings

M. Equi, H. Lievonen, Md R. Islam Khan, V. Mäkinen

arXiv 2026, under submission

Thank you! Questions?